

SyriaTel Customer Churn Prediction

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Phase 3 Data Science Project

Overview & Business and Data Understanding

Overview

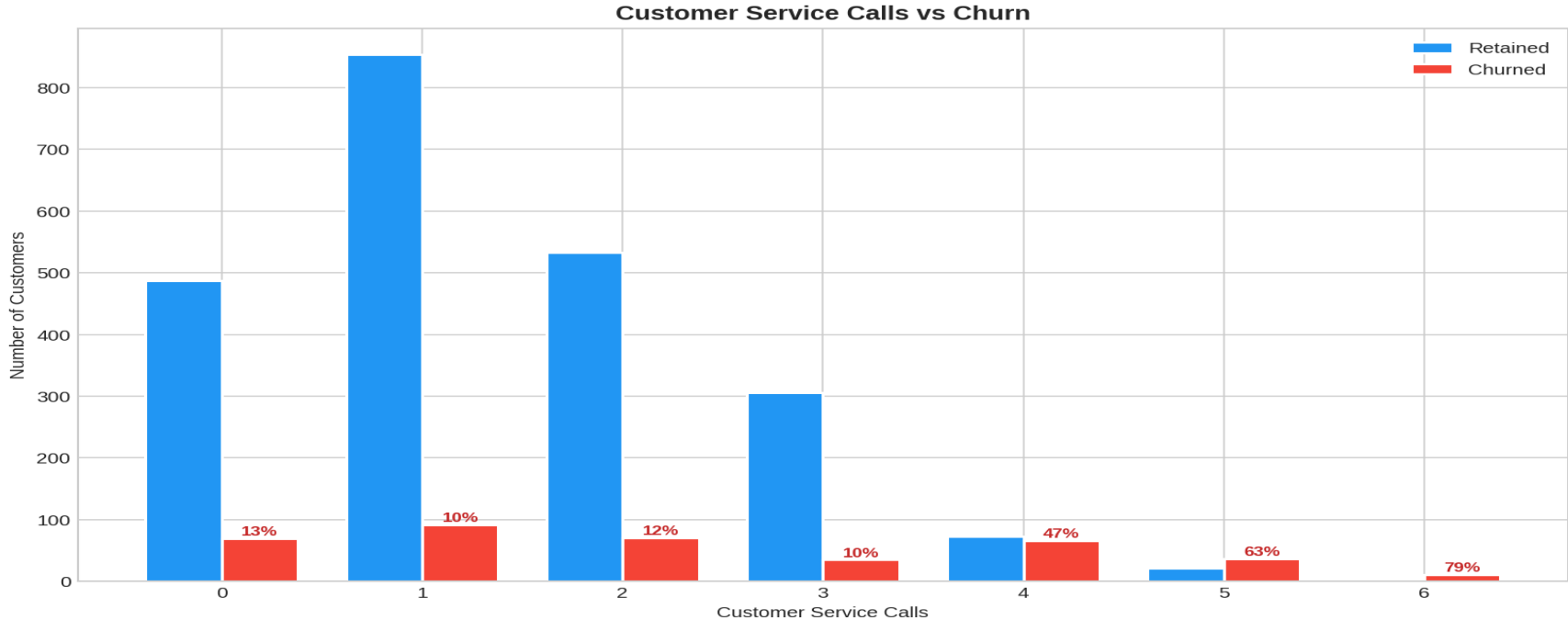
SyriaTel is a telecommunications company losing revenue to customer churn. This project builds a binary classification model to predict which customers are likely to leave , giving the retention team a chance to intervene before they do.

Business & Data Understanding

Stakeholder	SyriaTel's Customer Retention Team
Problem	Customers churn without warning, costing more to replace than retain
Dataset	3,333 customers · 21 features · No missing values
Churn Rate	14.5% — imbalanced dataset
Metric	Recall + F1-Score (churn class)

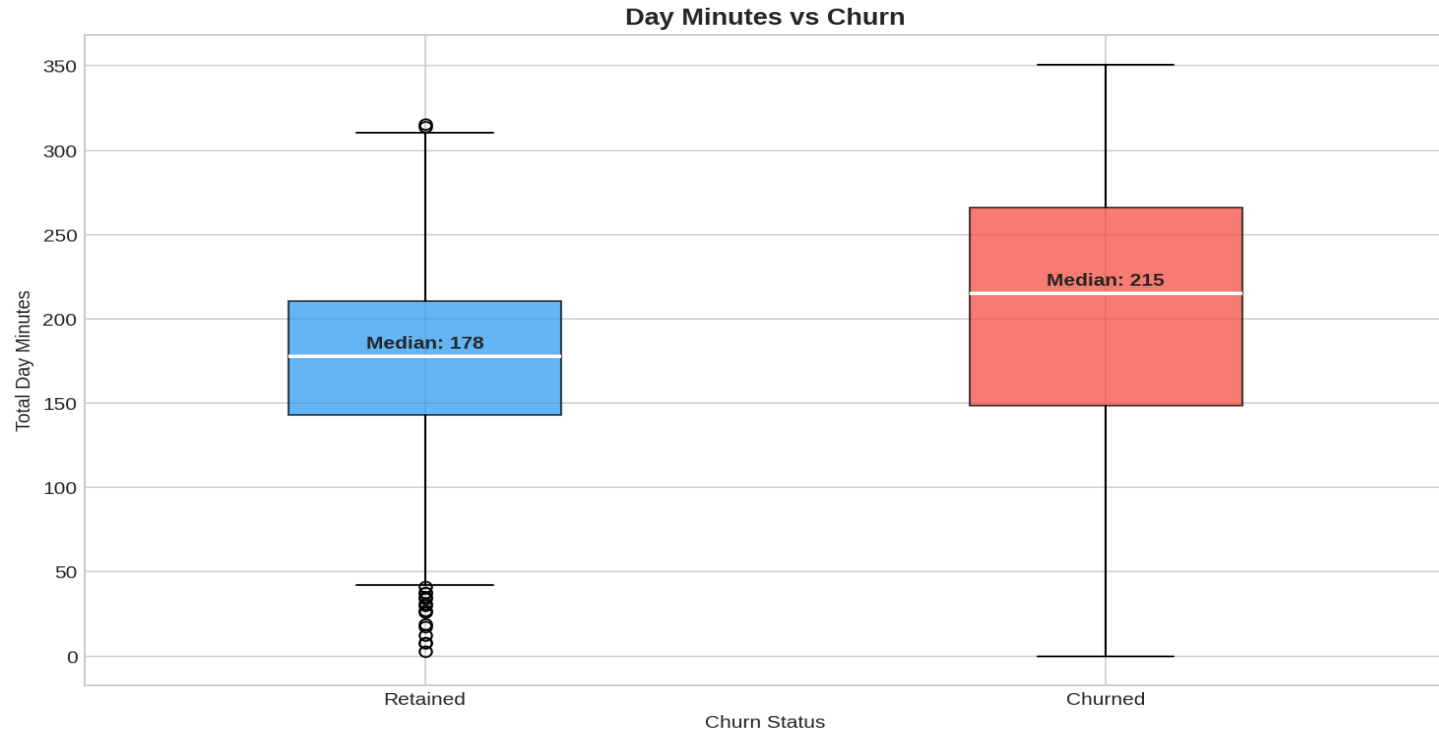
EDA

Q2 Analysis: Does poor customer service drive churn?



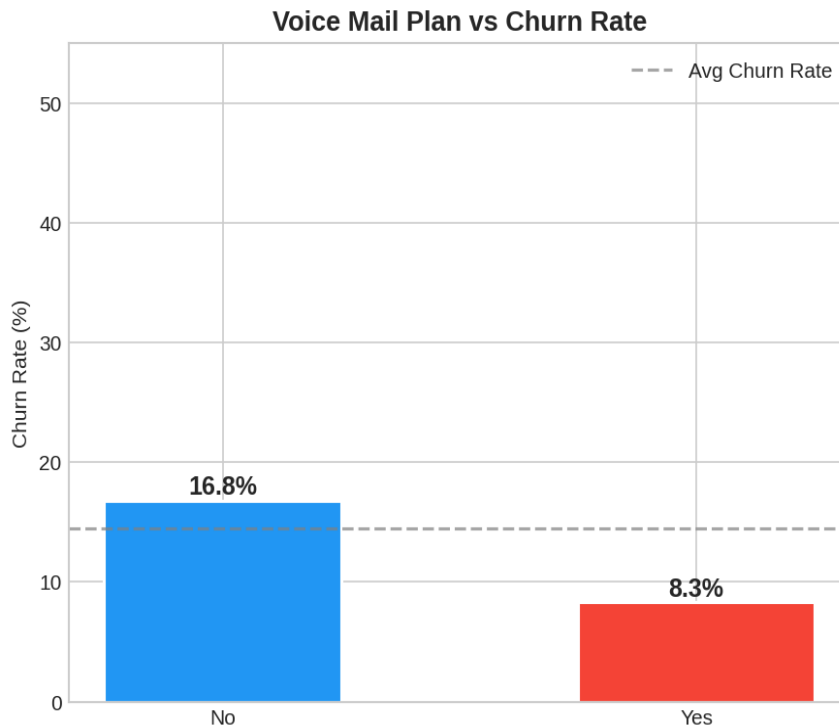
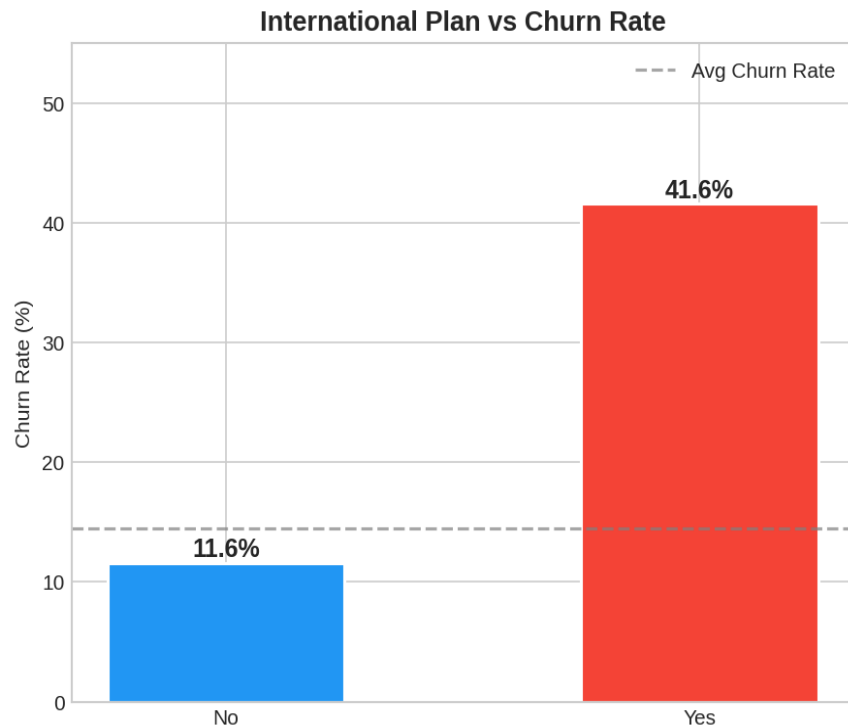
Customers who called support 4+ times churn at over 40% — nearly 3× the average rate.

Q3 Analysis: Are heavy users more likely to churn?



Churned customers average 215 day minutes vs 178 for retained customers — a 37-minute gap that means a consistently higher bill.

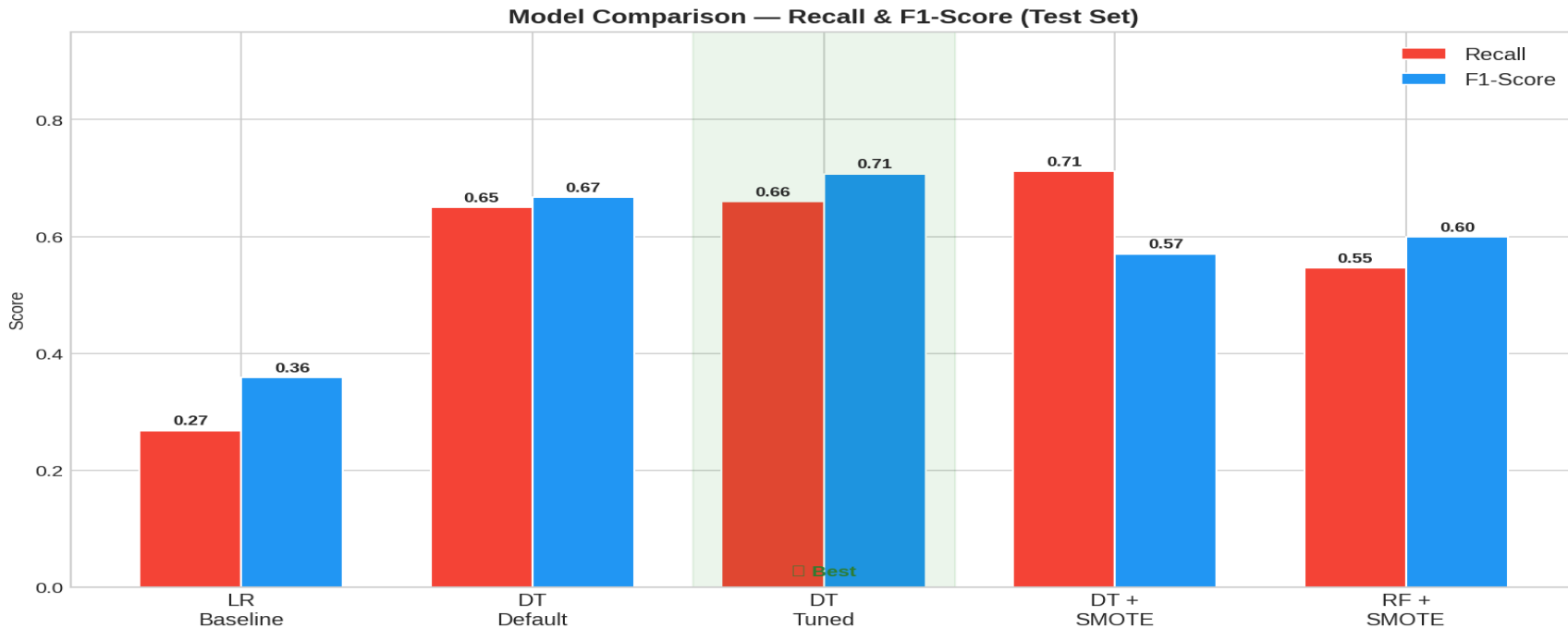
Q4 Analysis: Does the subscription plan predict churn?



International plan subscribers churn at 42% — nearly 4× the average. Voice mail subscribers churn at just 8.7%.

Modeling

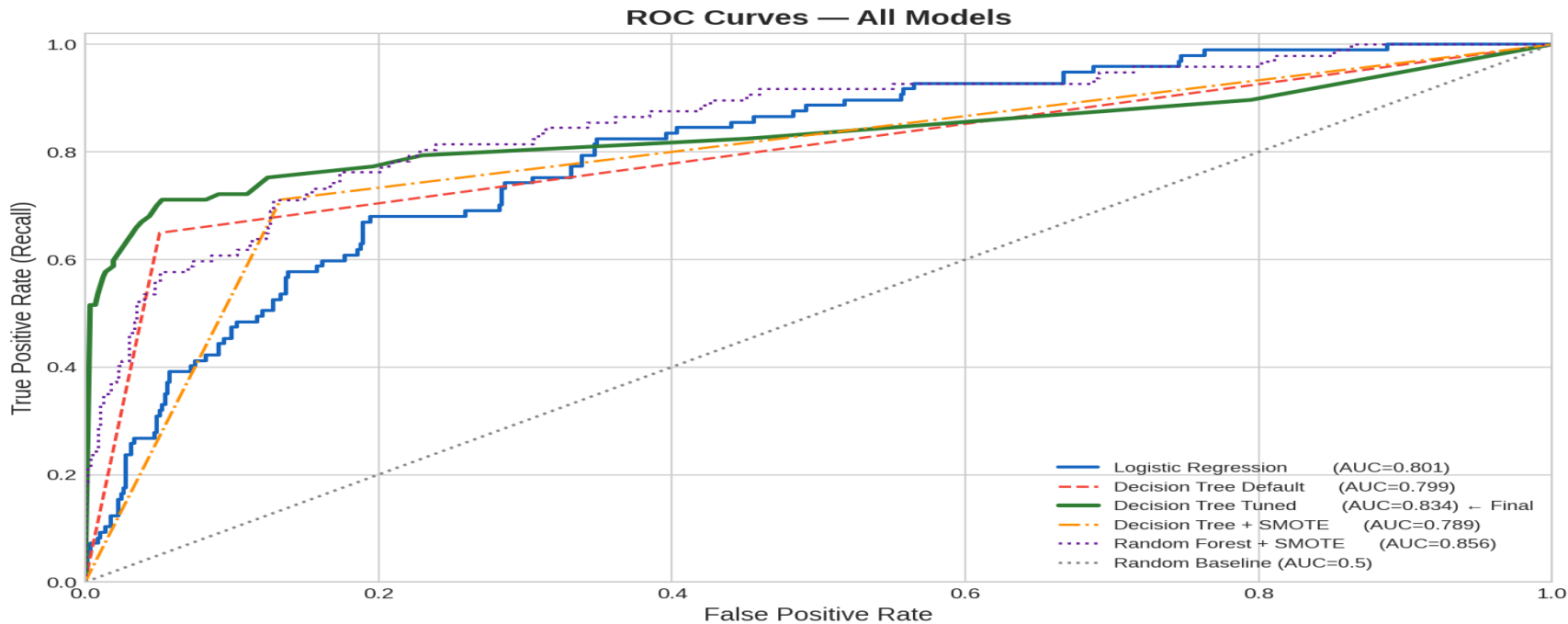
Which model performs best?



Five models tested. The Tuned Decision Tree achieves the highest F1-Score (0.707) which is the best balance between catching churners and minimising false alarms.

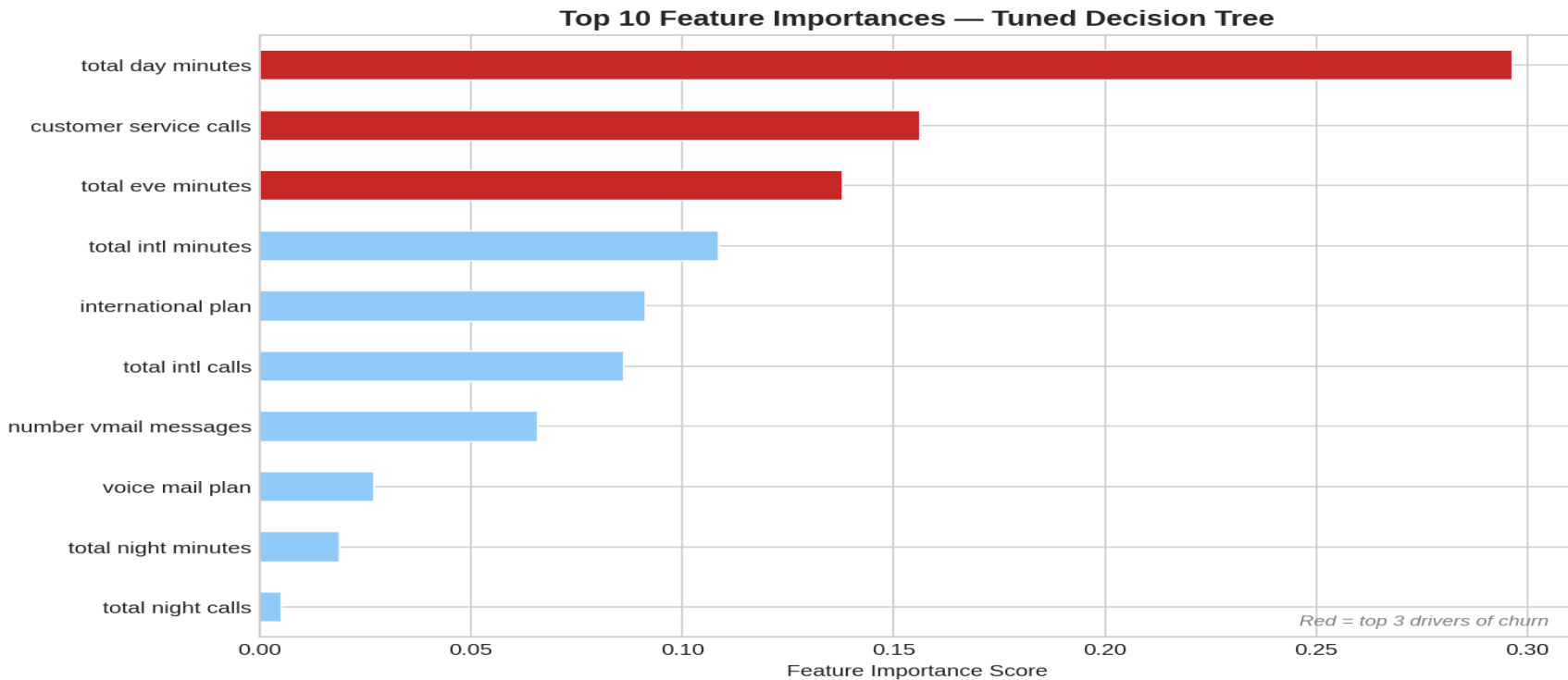
Evaluation

Evaluation 1: ROC Curves — All Models



The Tuned Decision Tree has the highest AUC (0.834), confirming it is the best model at distinguishing churners from non-churners across all thresholds.

Evaluation 2: What drives churn the most?



Total day minutes (0.30), customer service calls (0.16), and total eve minutes (0.14) are the top three drivers of churn.

Conclusion & Recommendations

Conclusion

The Tuned Decision Tree achieved the best balance of recall (0.660) and F1-score (0.707) across five model iterations. Total day minutes, customer service calls, and evening usage are the strongest predictors of churn — insights SyriaTel can act on immediately.

Recommendations

- Q1 — Flag customers after 3 service calls (churn jumps to 40%+ at call 4)
- Q2 — Audit the international plan urgently (42% churn rate)
- Q3 — Send billing alerts for customers using 215+ day minutes/day
- Q4 — Promote the voice mail plan to at-risk customers (8.7% churn rate)
- Q5 — Run the model monthly; focus on top 20% highest-risk customers

Next Steps

01

Collect richer data

Add actual bill amounts, complaint categories, and contract tenure to give the model stronger signals for predicting churn.

02

Retrain the model quarterly

Customer behaviour changes over time. The model should be updated regularly to stay accurate and relevant.

03

Explore advanced techniques

Try XGBoost or a tuned Random Forest without SMOTE to see if further performance gains are possible beyond the current F1 of 0.707.

Thank You

Questions & Discussion

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Model: Tuned Decision Tree | F1: 0.707 | AUC: 0.834

Data: SyriaTel Churn · Kaggle